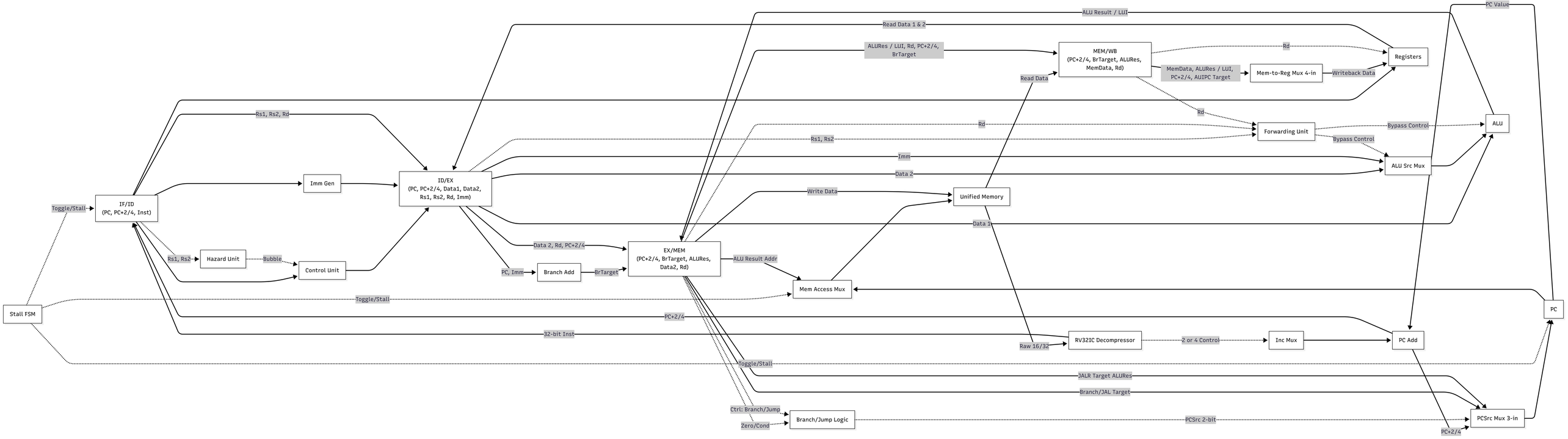
ALU Result / LUI
PC Value
Read Data 1 & 2
ALURes / LUI, Rd, PC+2/4, BrTarget
MEM/WB (PC+2/4, BrTarget, ALURes, MemData, Rd)
Rd
Registers
MemData, ALURes / LUI, PC+2/4, AUIPC Target
Mem-to-Reg Mux 4-in
Writeback Data
Read Data
Rs1, Rs2, Rd
Rd
Forwarding Unit
Bypass Control
ALU
Rs1, Rs2
Bypass Control
Imm
ALU Src Mux
Imm Gen
ID/EX (PC, PC+2/4, Data1, Data2, Rs1, Rs2, Rd, Imm)
Data 2
Write Data
Unified Memory
Toggle/Stall
IF/ID (PC, PC+2/4, Inst)
Data 1
Data 2, Rd, PC+2/4
EX/MEM (PC+2/4, BrTarget, ALURes, Data2, Rd)
Rs1, Rs2
Hazard Unit
Bubble
Control Unit
PC, Imm
Branch Add
BrTarget
ALU Result Addr
Mem Access Mux
Toggle/Stall
Stall FSM
PC
PC+2/4
32-bit Inst
RV32IC Decompressor
2 or 4 Control
Inc Mux
PC Add
Raw 16/32
Toggle/Stall
JALR Target ALURes
Branch/JAL Target
Ctrl: Branch/Jump
Zero/Cond
Branch/Jump Logic
PCSrc 2-bit
PCSrc Mux 3-in
PC+2/4